RESEARCH BRIEF · NORTH GARDEN INTEGRATION

Eight Months of Wrong Turns

A full accounting of an attempt to build a local, character-consistent pipeline for an original LitRPG webcomic — what was tried, what broke, why, and the questions still unanswered.

Compiled 31 Aug 2026 **Hardware** RTX 5090 Laptop · 24 GB

Status Base model change in progress **Purpose** Input for parallel research

01

The project

> An original LitRPG serial — novel plus webcomic — set on a real 70-acre property in North Garden, Virginia. Two adult leads based on the author and his wife, with consent. A daughter character exists in the fiction but is deliberately excluded from all image-model training.

Working title *North Garden Integration*. The premise: a rural property becomes the anchor point of a system integration event; the two leads develop complementary abilities — hers additive and visible (storm, root, growth), his subtractive (void, nullification, absence). A related JRPG project shares the setting.

The deliverable target is a **full chapter**: 50–90 panels, rendered, lettered, and correctable panel by panel. Not single hero images — those have been solved for months. The unsolved thing is *volume with consistency*.

WHY THIS IS THE HARD CASE

A survey of 19 published AI comics found the medium clusters in sci-fi and horror precisely because those genres tolerate atmospheric, single-character, face-obscured imagery, and that it struggles with "complex battle scenes or quieter slice-of-life stories." North Garden is a two-hander domestic drama with recurring interior sets and faces in nearly every panel. That is the documented worst case for this technology, and it explains a great deal of the difficulty below.

02

Constraints

HARD

- **Child safety.** The daughter's reference photos are isolated and excluded from every training path. No likeness model of a minor will be built. This is not negotiable and not up for optimization.
- **Likeness.** Two real, consenting adults. A LoRA is a durable artifact that generates new images of a real person's face. This constrains which hosted services are usable — see below.
- **Local GPU is shared.** The author games on the same machine; VRAM cannot be permanently consumed.
- **No new accounts, no unapproved spend.**

LIKENESS POLICY ACROSS VENDORS

This turned out to matter more than price. Cost for a full chapter through any commercial API is $5–30 — negligible. Policy is the real filter.

VENDOR	CONSENTED LIKENESS	PRACTICAL READ

Black Forest Labs (FLUX)	Explicitly contemplated — user warrants consent obtained	Only policy-safe hosted option. But prohibits training other models on outputs.
OpenAI	Permitted on paper	Deployed filters aggressive on identifiable people; no consent-attestation path. Legally fine, operationally unreliable.
Google	No carve-out	Jan 2026 tightening made refusals common. Highest policy risk.
ByteDance / Seedream	Blocks real faces at model layer	Unusable.
Adobe Firefly	Indemnity *excludes* real-person prompts and user reference images	Every exclusion describes this workflow.

Working conclusion: **identity stays local.** Hosted models, if used at all, serve as a staging or correction layer, with a final local pass to re-assert identity.

03

Current stack

Everything runs on the author's Windows machine, driven over ComfyUI's REST API from Python — no manual GUI steps, no batch files. The cloud agent environment has no GPU and no route to any model API, so the local machine is the only generation path.

LAYER	COMPONENT	STATE
Base model	Anima v1.1 aesthetic (Cosmos-Predict2-2B) + Qwen3-0.6B TE + Qwen-Image VAE	BEING REPLACED
Character identity	Two rank-16 LoRAs, triggers `s0rn` / `sgrd`, 27 and 31 images, 1500/1600 steps	WORKING
Trainer	AI-Toolkit, Python 3.12, torch cu128, ~40 min per LoRA	WORKING

Canon	`canon.py` — frozen description strings, negative sets, day-zero wardrobe variants	WORKING
Lettering	`strip/kit.py` — balloons, captions, SFX, System UI panels, page assembly, all in code	WORKING, UNUSED ON REAL ART
Stage geometry	`stage.py` — horizon/camera-height perspective sizing, occluder polygons, calibration overlay	WORKS, WRONG APPROACH
Effects	`voidfx.py` — code-drawn void: masking, radial space-bend, black-glass planes, falling darkness	WORKING

NOTABLE

Programmatic lettering is *unusual* among AI comic makers — the documented ones use Photoshop, Clip Studio, even Keynote — but it is standard practice in scanlation tooling (BallonsTranslator, manga-image-translator) and has an academic literature on balloon placement. At chapter scale this is an advantage, not a risk. Rewriting a line reflows automatically; a global style change costs nothing.

04

Attempt log

In order, because the order is the argument — each attempt was caused by the failure of the one before it. Every entry names what was actually wrong, not just that it looked bad.

01

Procedural vector art

Composited flat shapes in code as a stand-in for real art.

Rejected outright. "I didn't want it in this silly cutout style." Correct call — it was presented as progress when it was scaffolding. Pivoted to real diffusion.

02 Prompt-only character design

Catalogue sheets, six options per character, multiple art styles, iterated over many rounds.

Good for exploration, useless for consistency. No identity control whatsoever. Every sheet produced a different person. This is what motivated LoRA training.

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03 Character LoRAs

Rank 16 deliberately low, so the model learns identity rather than photographic skin texture from a photo dataset. Captions written `photo of <trigger>` so photographic-ness attaches to a token omitted at inference.

Worked. Verified by a controlled strength sweep — identical seed and prompt, only strength varying. Identity asserts from ~0.6.

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04 Two LoRAs in one frame

Both characters, both LoRAs loaded, single generation.

Failed structurally. Identity bleed, blended faces, and — with a posed prompt — uncanny stock-couple imagery. Described by the author as "absolutely frightening."

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05 Regional conditioning

`ConditioningSetMask` confining each trigger word to its own half of the frame. Core ComfyUI nodes, no install needed.

Failed, and the reason is the important part. A LoRA patches the *model*; the model is not regional. Text was regioned; both LoRAs still shaped every pixel, and one dominated globally. The male lead was absent from all eight attempts.

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06 Composite separately-generated figures

Characters generated alone on flat grey plates, keyed out via corner-sampled background masking, graded to room light, contact-shadowed, ink-edged, placed on separately generated rooms.

Best result so far, with a hard ceiling. Both characters reliably present and distinct. But a figure generated at eye level and pasted into a low-angle room is drawn in the *wrong perspective*, and no scaling corrects that.

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07 Perspective-derived scale

07 **Perspective-derived scale**

Figure size computed from room geometry rather than chosen: with camera at eye height E, an object of height E has its top on the horizon, giving `px_h = (y_feet - horizon) × (h / E)`. Camera height solved from a known object — a 0.75 m kitchen table. Plus occluder polygons so foreground furniture redraws over figures.

`**Fixed scale and the sitting-on-tables bug; introduced clipping.** Still compositing, so still fundamentally limited.`

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08 **Base model change — in progress**

Moving to Illustrious (SDXL family) for ControlNet, per-region LoRA via Impact Pack, IPAdapter, ~10× speed, and an open licence.

`**Underway.** Downloads and node installs running; LoRAs to be retrained.`

05

Root causes

> The durable findings. Several of these were expensive to learn and are not obvious.

THE BIG ONE

A 0.6B text encoder cannot parse compositional language. Anima's Qwen3-0.6B encoder operates effectively in Danbooru-tag space. Instructions like "she is a half step ahead, he is turned checking behind them" are not understood as spatial relations. Round after round of prompt tuning was spent fighting a model architecturally incapable of using the wording. Every staging failure in this project traces here.

Prompting

- **Naming an animal for scale makes the model draw that animal.** "Deer-sized" drew a deer; "the size of a bull" drew a bull, ears and all. Twice. Scale must be given against human height, never by species.
- **Arrangements produce photo shoots.** "Standing shoulder to shoulder facing the viewer" is an engagement-portrait instruction. Give two people no action and a camera to look at,

and the model composes the only thing that fits: a posed couple photo. Every panel needs an event.

- **Camera vocabulary summons photographs.** "Shallow depth of field", "cinematic", "dramatic rim light" fight style tokens and win. Drawing vocabulary — cel shading, ink outline, flat colour, screentone — is required instead.
- **Hammering absence flattens the image.** Background prompts leading with "empty room, no people, nobody" plus heavy person negatives produced dead, flat rooms. The model spends its capacity not-drawing. A rich description of the place plus a short negative produces far better environments.
- **Negatives leak into positives.** Writing "too many joints, extra limbs" in a prompt containing people produced limb chaos.
- **Never write "realistic proportions" or "mature adult face"** — they force photorealism against every style token.
- **Dilution kills regions.** Prepending a full room description to each region prompt made the character a minority of their own region; stripping it entirely collapsed the base. Region text must carry only what differs.

Training

- Photo datasets on an anime base drag output photoreal — but this is **strength- and framing-dependent**, not baked in. It appears at close range where the face dominates the frame, and disappears at 0.5–0.65 strength with drawing vocabulary. The documented fallback (retrain on pre-stylised data) was never needed.
- Different strengths for different shot types: ~0.8 wide, 0.5–0.65 close, ~0.6 with two characters.
- Trainer sample images are *not* representative — different sampler, no curated negatives. Judge a LoRA through the real pipeline only.

Canon and continuity

- **Character strings must be frozen constants in fixed slot order.** Text encoders are order-sensitive; rewording an identical description produces a different face. Hand-writing a character description per prompt was the direct cause of one lead drifting between images.

- **Canon has a timeline.** The description strings encoded the leads' post-integration look — tattoos, harness, axes — so chapter one dressed them for a fight during a conversation about propane. Wardrobe needs era variants.
- **Character plates invent props.** A figure plate that generates its own lamp lands in a room that already has one. Poses must describe the body only and let the room supply furniture.

Composition mechanics

- Three things make a composite read as a drawing rather than a collage: **grade** the figure toward the room's ambient light, a **contact shadow** so it doesn't hover, and a redrawn **ink edge** so the cut reads as linework. A heavy ink style is what makes compositing viable at all; a painted style would expose every seam.
- Latent and pixel blending are off-manifold by construction, and VAE round-trips accumulate colour drift. Compositing has a mathematical ceiling, not just an aesthetic one.

Environment and tooling

- Processes launched from an agent tool call are killed with that call's process tree. Long jobs must be detached.
- Anima's latent carries a third spatial axis, so ComfyUI's percentage-area conditioning throws `IndexError`; the mask path takes a different branch and works.
- `/object_info` in full resets the connection; query one node at a time.

06

Open problems

PROBLEM	STATE	NOTES
Multi-character panels	PARTIAL	Compositing works; perspective mismatch and clipping remain. Per-region LoRA on Illustrious is the untested fix.
Figure/room	OPEN	Unfixable by compositing. Requires generating in-scene

Figure/room perspective mismatch	OPEN	Unfixable by compositing. Requires generating in-scene.
Set continuity across a chapter	OPEN	Reusing one room plate works; no method yet for the same room from a new angle.
Palette runaway	PARTIAL	Amber accent overwhelms the intended slate; negatives only partly hold it.
Panel-level correction loop	OPEN	Instruction-edit models identified as the answer; not yet built.
Drift detection at chapter scale	OPEN	No automated audit exists. Identified as highest-value custom tooling.
Art style not finalised	OPEN	Author has not down-selected. Blocks style-LoRA training.
Land reference photography	OPEN	The real property is the setting; no plates from it yet.
Void power reading	PARTIAL	Code-drawn effect works. Underlying asymmetry: her power is additive and drawable, his is subtractive and "nothing" is not a picture. Best fix is to make the absence *act*.

07

Research questions

> The payload of this document. Ordered by expected value, not by difficulty. Where a hypothesis exists it is stated so it can be attacked rather than rediscovered.

Q1 · Per-region LoRA

Does Impact Pack's `RegionalSampler` on an SDXL/Illustrious base actually deliver two distinct character identities in one frame, at production quality? What are the failure modes at region boundaries, and how does `base_only_steps` trade composition coherence against identity strength?

Hypothesis: this is the single change that unblocks two-character panels. Untested.

Q2 · Generate-in-scene vs composite

Compare, on identical staging: (a) composite cutouts, (b) sequential inpainting of each character into the finished room with only their LoRA loaded, (c) regional sampling in one pass. Measure perspective correctness, lighting integration, edge quality, and identity retention.

Hypothesis: (b) wins – the character inherits the room's real light and perspective because it is generated in place. Cost is serial passes per panel.

Q3 · 3D blockout as the staging layer

Does posing low-poly proxies in Blender and exporting depth + OpenPose + per-character segmentation masks eliminate the scale, occlusion and staging problems outright? What is the real per-panel cost once a set and two rigs exist? Does it survive stylisation, or does depth-conditioning flatten the art?

This would replace the entire hand-tuned perspective system with physical ground truth, and the segmentation masks come out pre-registered for Q1/Q2. A Blender integration is already available in this environment.

Q4 · Set continuity from new angles

How do you render the same room from a different camera and have it remain recognisably the same room? Candidates: 3D set built once; IPAdapter with the establishing plate as reference; a per-location LoRA; panoramic generation then crop.

Least-served problem found in the research. No first-hand account of anyone training an environment LoRA for a comic was located.

Q5 · Automated drift audit

Can a VLM reliably walk a rendered chapter, match characters across panels by persistent attributes, flag drift, and distinguish *intentional* narrative change from accidental drift? What's the false-positive rate, and what annotation format best drives an automated repair pass?

Every practitioner account names drift as the dominant failure and manual eyeballing as the dominant cost. Published work suggests this is tractable. Nothing off the shelf does it.

Q6 · Instruction-edit as the correction loop

How reliable are current instruction-edit models for comic panel repair — "change her expression to worried", "move him left", "remove the second lamp" — while preserving identity and style? Where do they silently degrade the rest of the frame? Can identity be re-asserted afterward with a low-denoise LoRA pass?

This is the mechanism for panel-by-panel correction at chapter scale. Directly requested.

Q7 · Style LoRA and the ceiling

Does a project style LoRA measurably improve chapter coherence over prompt tokens alone? How many images does it actually need? And is the counterintuitive finding true — that a *simpler*, flatter style has a higher quality ceiling because it hides model artefacts?

Bears directly on the still-open art-style decision.

Q8 · The subtractive power problem

An art-direction question, not a technical one: how have comics historically depicted absence, negation and nullification as *power*? Absence is not a picture; the world's reaction to it is. What is the visual vocabulary?

Six rounds of prompting failed to make "holes in his silhouette" read as anything but a rendering error. The eventual fix was to draw the effect in code — but the design question stands.

Q9 · Panel-record architecture

What is the right data model for a chapter where every stage is addressable by panel id — so "fix panel 34" never touches panel 33? Fields, state machine, revision history, candidate management, and how a lettering pass consumes only approved panels.

The connective tissue that makes everything above usable at chapter scale. A few hundred lines of glue over tools that all exist.

Q10 · Honest economics

What does a 70-panel chapter actually cost in hours for a solo operator, and how much of that is the hard tail? Existing estimates (15–30 hours, 3–5× faster than traditional) are synthesised from two data points and need calibration against a real chapter.

Determines whether a serial is sustainable at all.

08

Unverified

Stated plainly so no one downstream treats these as established.

- No verified account of a **serialized** AI webtoon at weekly cadence was found. Every solid case study is a one-off book. Recurring sets and cast are less trodden than search volume suggests.
- Whether experimental Anima ControlNet weights (trained on Base v1.0) work on the Aesthetic v1.1 build – no source either way.
- Whether Impact Pack's regional sampling works on Anima specifically – architecturally plausible, unstated anywhere.
- Consistency percentages cited in the literature come from a single author with no published methodology.
- Time estimates are a synthesis of two real data points plus traditional baselines, not measurement.
- Reddit and several gated model-community article sources were inaccessible to the research passes – likely where the richest practitioner detail lives. Worth a direct human look.